MIL-C-58021B(AV)
5 March 1981
Superseding
MIL-C-58021A(AV)
18 August 1969

MILITARY SPECIFICATION

CABINETS AND BENCH TOPS

1. SCOPE

1.1 Scope.--This specification covers steel storage cabinets and wood bench tops.

1.2 Classification.--Cabinets shall be of the following types, and bench tops of the following sizes, as specified:

1.2.1 Types of cabinets.--

Type I	Drawer storage.
Type II	Open bin storage.
Type III	Supplement storage.
Type IV	Paint storage.

1.2.2 Sizes of bench tops.--

Size A	84 by 30 by 1-3/4 inches.
Size B	63 by 30 by 1-3/4 inches.
Size C	42 by 30 by 1-3/4 inches.
Size D	21 by 30 by 1-3/4 inches.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

FF-N-836	Nut, Plain: (Hexagon, Square, Cap and Welding) Nut, Slotted, and Castellated, Hexagon
FF-S-92	Screws, Machine: Slotted, Cross-Recessed or Hexagon Head
FF-W-84	Washers, Lock (Spring)

FSC 4920

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specification shall take precedence.

FF-W-92	Washers, Metal, Flat (Plain)
MM-L-736	Lumber and Timber: Hardwood
QQ-S-624	Steel Bar, Alloy, Hot Rolled and Cold Finished (General Purpose)
QQ-S-741	Steel, Carbon: Structural Shapes, Plates, and Bars
TT-P-1757	Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity
TT-W-572	Wood-Preservative; Water Repellent
MMM-A-125	Adhesive, Casein-Type, Water and Mold Resistant
MMM-A-188	Adhesive, Urea-Resin-Type (Liquid and Powder)
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-T-60	Tape: Pressure-Sensitive Adhesive; Waterproof, for Packaging

MILITARY

MIL-P-116	Preservation, Methods of
MIL-T-704	Treatment and Painting of Material

STANDARDS

FEDERAL

FED-STD-66	Steel: Chemical Composition and Hardenability
FED-STD-595	Colors

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Equipment
MIL-STD-1188	Commercial Packaging of Supplies and Equipment
MS20470	Rivet, Solid -- Universal Head, Aluminum and Aluminum Alloy

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DRAWINGS

DEPARTMENT OF THE ARMY

AA-17A-0002	Base Cabinet Assy - Drawer Storage Type I
AA-17A-0002-1	Base Cabinet Frame Weldment Type I
AA-17A-0002-2	Base Cabinet Details Type I
AA-17A-0002-3	Slide Angle Weldment Type I
AA-17A-0002-4	Drawer Assembly - 3-1/4" Type I
AA-17A-0002-5	Drawer Assembly - 7-3/16" Type I
AA-17A-0002-6	Drawer Assembly - 11-1/8" Type I
AA-17A-0003	Base Cabinet Assy - Open Bin Storage Type II
AA-17A-0003-1	Details, Open Bin Storage Cabinet Type II
AA-17A-0003-2	Open Bin Storage Cabinet Type II Details
AA-17A-0004	Supplement Cabinet Assy Type III
AA-17A-0004-1	Supplement Cabinet Assy, Details Type III
AA-17A-0004-2	Supplement Cabinet Assy, Details Type III
AA-17A-0004-3	Drawer Assembly - 2" Type III
AA-17A-0004-4	Details, Supplement Cabinet Assy, Type III
AA-17A-0005	Bench Tops - Assy & Details Size A, B, C & D
AA-17A-0020-1	Paint Storage Cabinet Type IV
AA-17A-0020-2	Paint Storage Cabinet Details Type IV

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.--The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 Screw-Thread Standards for Federal Services

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(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

AMERICAN WELDING SOCIETY

WELDING HANDBOOK

(Application for copies should be addressed to the American Welding Society, 2501 N.W. 7th Street, Miami, FL 33125.)

3. REQUIREMENTS

3.1 Design Characteristics.--Cabinets shall be of four types and designed to provide drawer and open bin storage. Types I, II, and III (see 1.2.1) shall be designed for stacking where required or may be fitted with bench tops for use as a working space. These cabinets shall be provided with means for locking on both sides, and if one side is locked, to be accessible from the other side. Type IV cabinet shall be accessible from one side only and be provided with means for locking. Cabinets and bench tops shall be fabricated as specified herein and shown on the drawings in 2.1. If a change in design, or additional components, are required to conform to this specification, emphasis shall be placed on simplicity of design, lightweight, and ease of fabrication by mass production. If there is any conflict between the drawings and the requirements of this specification, the latter shall govern.

3.2 Materials.--Materials shall be as specified herein and as shown on the applicable drawings in 2.1. Material not definitely specified shall be of standard commercial quality for this type of equipment.

3.2.1 Steel.--

3.2.1.1 Sheet.--Sheet steel used in the construction of the cabinets, where spot welding is specified, shall conform to No.1010 of FED-STD-66.

3.2.1.2 Structural.--Brackets, frames, and other shapes shall be fabricated of steel conforming to Type II of QQ-S-741.

3.2.2 Hardwood.--Bench tops shall be of kiln-dried first grade oak conforming to MM-L-736.

3.2.3 Glue.--Glue used in the construction of the laminated bench tops shall be a water-resistant casein or urea-resin-type glue in conformance with the applicable requirements of MMM-A-125 or MMM-A-188.

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3.3 Fastenings.--All screws, bolts, fastener assemblies, and other similar devices shall be installed with a positive means of locking to prevent loosening due to vibration. Locking shall be by cotter pins, locknuts, castellated nuts, positive lockwashers, or any other locking method approved by the contracting officer. Swedging, peening, or staking of threads, on parts which are subject to removal for disassembly, shall not be permitted. Threaded fasteners shall have approximately two threads showing when completely tightened.

3.3.1 Machine screws.--Machine screws conforming to FF-S-92 shall be of steel conforming to QQ-S-624 and of the type as specified on the applicable drawings in 2.1.

3.3.2 Bolts.--Through bolts shall be of steel conforming to Type I of QQ-S-741, and shall be of the thread series and size as specified on Drawing AA-17A-0005.

3.3.3 Nuts.--Nuts shall conform to Type I of FF-N-836 and shall be of the size and thread series specified on the applicable drawings in 2.1.

3.3.4 Weshers.--Flatwashers shall conform to Type A, grade I, Class A of FF-W-92 and lockwashers shall conform to Class A, style 2 of FF-W-84. Washers shall be of the size specified on the applicable drawings in 2.1.

3.3.5 Rivets.--Rivets shall conform to MS20470 and shall be of the type and size specified on the applicable drawings in 2.1.

3.4 Construction.--

3.4.1 Welding.--The surface of all parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Unless otherwise specified, or shown on the drawings, welds shall develop the strength as required for the parts connected. All welding shall be in accordance with the latest applicable codes of the American Welding Society as contained in the Welding Handbook. Welders shall be qualified in accordance with the applicable tests specified in the Welding Handbook. Any welder shall be required to retake, and satisfactorily pass this test, when, in the opinion of the contracting officer, there is a reasonable doubt of his proficiency.

3.4.2 Standard parts.--AN or MS standard parts shall be used wherever they are suitable for the purpose and shall be identified by their part numbers. Commercial utility parts such as screws, bolts, nuts, and cotter pins may be used provided they have suitable properties and are capable of being replaced by AN or MS standard parts without alteration. In applications for which no suitable corresponding AN or MS part is in

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effect on date of invitation for bids, commercial parts may be used provided they conform to this specification.

3.4.3 Threaded parts.—Threaded parts shall conform to Handbook H28. Thread series shall be as specified on the applicable drawings listed in 2.1.

3.4.4 Interchangeability of parts.--Components shall be interchangeable (without requiring modification for replacement) with similar components furnished under the same contract or order.

3.4.5 Riveted connections.--Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall completely fill the holes. Surfaces to be riveted shall be painted with primer zinc-chromate conforming to TT-P-1757 prior to assembly. Rivet heads when not countersunk or flattened shall be of buttonhead shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.4.6 Adjustments.--All parts subject to adjustment or replacement by operating or maintenance personnel shall be readily accessible. Adjusting facilities, where provided, shall operate smoothly and without interference and shall adequately maintain such adjustments under all conditions of service. Designs which render servicing, adjustment, or replacement unduly difficult under field conditions will not be accepted.

3.4.7 Equivalent products.--Where a manufacturer's products are shown on the drawings referenced in 2.1, equivalent products may be substituted in the preparation of the manufacturer's working drawings. However, after approval of the working drawings by the contracting officer, products specified thereon shall be furnished without substitution.

3.5 Cabinets.--The four types of cabinets (see 1.2.1) shall be as specified herein and as shown on the drawings in 2.1.

3.5.1 Type I.--Type I cabinet shall consist of two vertical rows of six drawers. Assembly and details shall conform to Drawings AA-17A-0002 through AA-17A-0002-6. Cabinet shall be complete with locking bars and hardware on both sides.

3.5.2 Type II.--Type II cabinet shall consist of two open bins with adjustable shelves, a removable wire enclosure on both sides, and provisions for locking each side separately. Assembly and details shall conform to Drawings AA-17A-0003 through AA-17A-0003-2.

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3.5.3 Type III.—Type III cabinet shall consist of 10 drawers and an open bin on top and the necessary hardware and locking bars on both sides. Assembly and details shall conform to Drawings AA-17A-0004 through AA-17A-0004-4.

3.5.4 Type IV.--Type IV cabinet shall consist of three shelves divided into nine compartments, with three removable wire enclosures in front, and shall be complete with locking facilities. Assembly and details shall conform to Drawings AA-17A-0020-1 and AA-17A-0020-2.

3.6 Bench tops.--Bench tops shall be of four sizes (see 1.2.2) and of laminated oak construction (see 3.2.2), glued together, and secured with through bolts as shown on Drawing AA-17A-0005.

3.7 Treatment and painting.--

3.7.1 Cabinets.--All surfaces, interior and exterior, shall be treated and painted in accordance with the applicable requirements for type C of MIL-T-704. The finish coat shall be olive drab conforming to No. 24087 of FED-STD-595.

3.7.2 Bench tops.--The working surface of the bench tops, after being burred and sanded smooth, shall be given two coats of hot linseed oil with a time lapse between coats to permit drying. The edges and underside of the bench tops shall be given two coats of water-repellent preservative solution conforming to TT-W-572. The solution shall be colored olive drab, No. 24087 of MIL-STD-595.

3.8 Marking.--Cabinets and bench tops shall be marked individually for identification in conformance with MIL-STD-130.

3.9 Workmanship.--Workmanship on the cabinets and bench tops, including all parts and accessories, shall be first class in every respect, and free from defects that could affect appearance and satisfactory performance. Particular attention shall be given to accuracy of dimensions, neatness, marking of parts and assemblies, welding, painting, riveting, machine-screw assemblies, and freedom from burrs and sharp edges.

3.9.1 Steel fabrication.--Steel used in the fabrication of the equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true. Flame cutting, by a tip suitable for the thickness of metal, may be employed instead of shearing or sawing. Burned surfaces of flame-cut material shall be ground or machined sufficiently to remove ash and cooling checks.

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3.9.2 Threaded connections.--All bolts, nuts, and screws shall be tight. "Tight" means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw, bolt, or threads.

3.9.3 Cleaning.--After final assembly, items shall be thoroughly cleaned of loose, spattered, or excess weld, metal chips, and other foreign material. Burrs and sharp edges shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.--Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conformed to prescribed requirements.

4.2 Quality conformance inspection.--Each cabinet and bench top shall be inspected to determine:

4.2.1 Functional.--

a. Presence of all drawers, bins, and handles.
b. Completeness and correctness of assemblies and fastenings.
c. Operability of all drawers and latches.
d. Identification and other required marking.

4.2.2 Workmanship.--

a. Suitability of fits at joints, seams, and edges.
b. Treatment and painting.
c. Absence of defects such as scale, lumps, runs, craze, thin spots, and holidays on the surfaces.
d. Metal finish as specified.

4.3 Inspection of preparation for delivery.--Preservation and packaging, packing, and marking shall be inspected prior to delivery to determine compliance with section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.--Cabinets and bench tops shall be packaged level A or commercial, as specified (see 6.2).

5.1.1 Level A.

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indicator set to 0. The ring gage shall be applied six times without a variation

5.1.1.1 Cleaning and drying.--Cabinets and bench tops shall be cleaned in accordance with process C-1 and dried in accordance with process D-1 or D-4 of MIL-P-116.

5.1.1.2 Preservation.--Latches and other operational surfaces shall be preserved with P-7 type preservative. All other unpainted metal surfaces shall be coated with P-1 type preservative, in accordance with MIL-P-116.

5.1.1.3 Unit Packaging.--Each unit shall be packaged Method I of MIL-P-116. If drain holes are not provided, drawers shall be inserted upside down to facilitate drainage of condensation. Cabinet tops and faces and bench tops shall be covered with sheets of fiberboard, fastened securely with PPP-T-60 tape, and cushioned to prevent scratching or marring of surface. Assembly instructions and drawings shall be packaged Method IC-1 of MIL-P-116 and secured to the basic unit.

5.1.2 Commercial.--Cabinets and bench tops shall be preserved and packaged in accordance with MIL-STD-1188.

5.2 Packing.--Cabinets and bench tops packaged per 5.1 shall be packed level A, B, or commercial, as specified (see 6.2).

5.2.1 Level A.--All components comprising one complete cabinet and bench top shall be blocked, braced, and packed in a PPP-B-585, Class 3 wirebound wood box, or a PPP-B-601, type overseas wood-cleated plywood box, or a PPP-B-621 nailed wood box. Closure shall be in accordance with the applicable box specification.

5.2.2 Level B.--Items shall be packed as for level A, except that PPP-B-591 wood-cleated fiberboard box, or PPP-B-621, Class 1 nailed wood box, or PPP-B-585, Class 1 wirebound box, or PPP-B-601, type domestic cleated plywood box may be used.

5.2.3 Commercial.--Packages shall be overpacked in accordance with MIL-STD-1188.

5.3 Markings.--In addition to any special markings required by the contract or order, packages shall be marked in accordance with MIL-STD-129 or MIL-STD-1188, as specified (see 6.2).

6. NOTES

6.1 Intended use.--The cabinets and bench tops are components of shop sets intended for use in mobile aircraft maintenance vans.

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6.2 Ordering data.--Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Types of cabinets and sizes of bench tops required (see 1.2).
c. Level of packaging and packing required (see 5.1 and 5.2).
d. Markings required (see 5.3).

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Preparing activity:

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